

CHAHAK GUPTA

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EDUCATION

University of Illinois Urbana-Champaign — James Scholar

Dual Degree: Bachelor of Science in Computer Science & Bachelor of Science in Statistics (LAS)

Cumulative GPA: 3.91/4.0, Dean's List

- **Coursework:** Data Structures, Algorithms & Models of Computation, Programming Languages & Compilers, Database Systems, Artificial Intelligence, Numerical Methods, Computer Architecture, Statistical Analysis/Modeling
- **Organizations & Activities:** NOBE Consulting Club (Tech Dev); Alpha Omega Epsilon (Technology Director)

EXPERIENCE

Software Engineering Intern — *Alarm.com, Inc.*

Jun 2026 - Present

- Reduced **WebRTC camera stream load time 25% (1,620ms)** by introducing **camera stream preloading**, initiating connections ahead of user request to bypass heavy startup overhead (DNS lookups, ICE/SDP handshakes, codec init).
- Built a real-time progress UI in Kotlin / Jetpack Compose for a **security interface (35,000+ active users)**, mapping backend stream-connection callbacks to UI via a custom listener/enum mapper, and feature toggle gating with unit tests.

Undergraduate Research Assistant — *Computational Neuroscience Lab*

Dec 2025 – Present

- Modeled high-dimensional MRI data from **227 subjects** using Gaussian Process spatial smoothing to estimate regional neural activity, **improving signal consistency 15%** over baseline interpolation for more reliable downstream analysis.
- Built and cross-validated ML classifiers (Scikit-Learn) to infer latent neural states from MRI signals achieving AUC=0.78.
- Implemented regression and optimization pipelines to estimate parameters of neural systems, **reducing prediction error (RMSE) by 12%** compared to baseline models.

Software Engineering Team Tech Developer — *Cayden Cay Consulting*

Jan 2026 – Jun 2026

- Agile development of a full-stack React Native / Firebase / SQL app for local startup client using Claude Code, and directing feature scope and delivery for LMS, video streaming, and social feed modules serving **50+ users**.
- Architected backend authentication, role-based access control, and data models to support multi-feature scalability; **cut API response time 25%** via Firebase query and SQL schema optimization.

CS Research Scholar — *Systems Platform Research Group*

Aug 2025 – Jan 2026

- Designed and evaluated STRATUS, a CrewAI-based multi-agent SRE framework for automated cloud failure detection and mitigation, **reducing simulated incident response time by 30%**.
- Aggregated and analyzed distributed observability data (logs, traces, metrics) across 20+ microservices to enable LLM-driven fault localization and root-cause analysis, **improving fault detection precision by 18%**.

Software Engineering Intern — *Keybank*

May 2024 – Jul 2024

- Prototyped **retrieval-augmented generation (RAG) models** for banking applications, using LangChain, Vertex AI, and OpenAI APIs to benchmark tradeoffs in latency and cost across configurations.
- Developed Python pipelines integrating Pinecone to benchmark model accuracy and **reduce query latency by 30%**.

PROJECTS

Limit Order Book Engine

May 2026 – Present

- Engineered a low-latency **limit order book matching engine** in C++ using a red-black tree (std::map) for price-level indexing, supporting market and limit order types with FIFO price-time priority matching.
- Designed a lock-free queue feeding a single-threaded execution core, achieving **30–250ns average per-order latency** at 100k+ concurrent orders.

InternScout

May 2026 – Present

- Built an end-to-end internship pipeline scraping **Greenhouse and GitHub Jobs** (40+ postings/day), with hash-based dedup and config-driven role/location/visa filtering to eliminate noise.
- Developed a semantic ranking pipeline combining **sentence-transformer embeddings** with **Llama 3.3** scoring to match postings against my resume, surfacing only high-confidence fits.
- Automated real-time mobile alerts via **Pushover**, persisting job state in PostgreSQL to prevent duplicate notifications, saving **4 hours/week** of manual searching.

SKILLS

Languages & Frameworks: Go, Java, C++, Kotlin, Python, Javascript, React, Node.js, Android SDK, Firebase, R, SQL

Tools & Systems: Git, Docker, CI/CD, Agile, Linux, AWS, Azure, REST APIs, Kubernetes

Data Science & ML: Scikit-Learn, PyTorch, TensorFlow, Pandas, NumPy